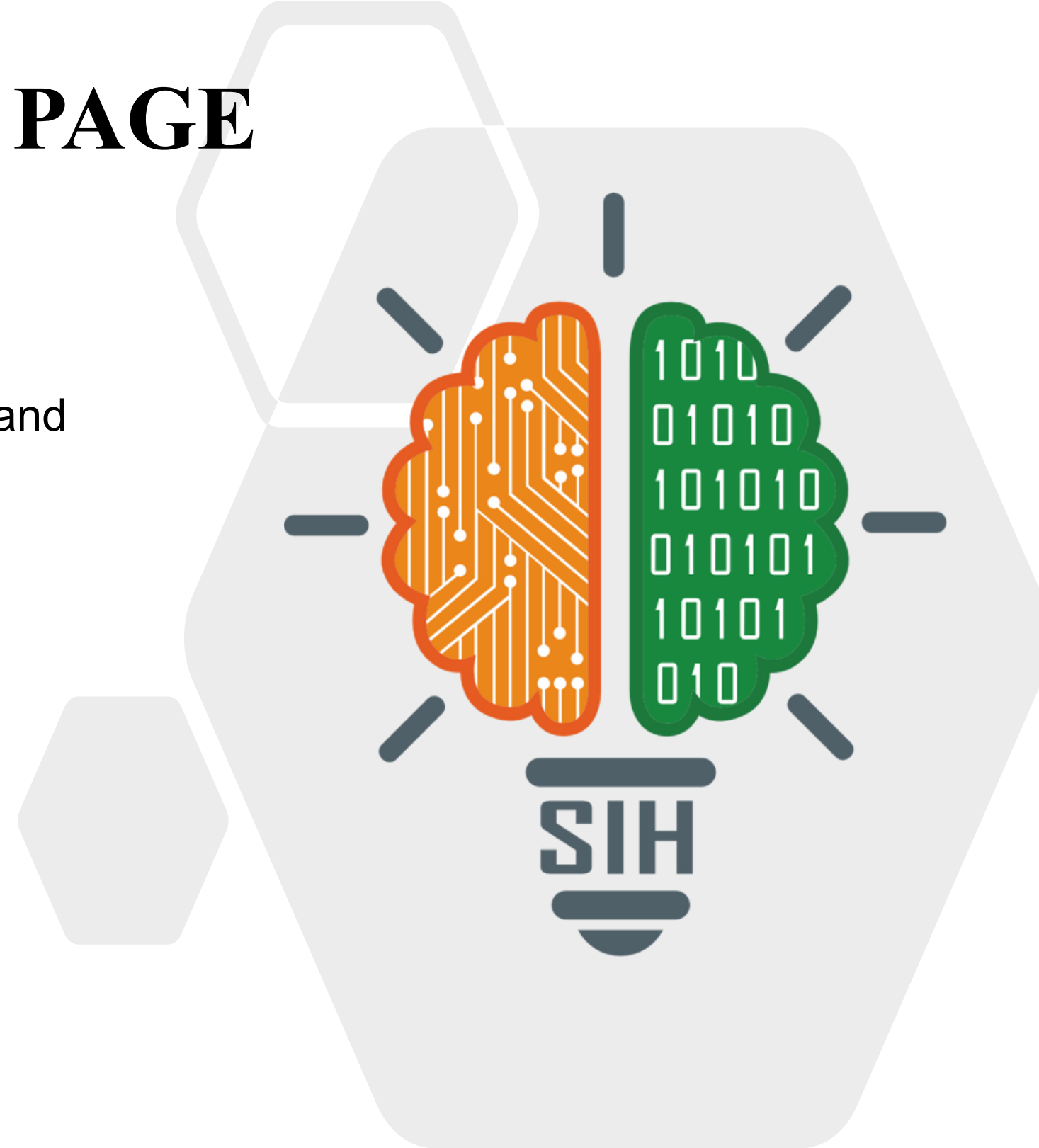


TITLE PAGE

- **Problem Statement ID** – SIH26037
- **Problem Statement Title** – Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme** – Smart Vehicles
- **PS Category** – Software
- **Team ID** – 161638
- **Team Name** – Ctrl Alt Elite



Adaptive Path Planning & Collision Avoidance for Unstructured Indian Roads



PROPOSED SOLUTION

Proposed Solution (Describe your Idea/Solution/Prototype) · Detailed explanation of the proposed solution

PathSense is an adaptive path-planning and collision-avoidance system for autonomous vehicles on unstructured Indian roads.

It reasons about **mixed traffic** — pedestrians, motorcycles, auto-rickshaws, cars, buses, trucks and animals — on roads with **missing or unclear lane markings**, informal merges, sudden direction changes, wrong-way movement and temporary obstacles.



concept illustration



PROBLEMS IT ADDRESSES

How it addresses the problem

01



Unmarked / Unstructured Roads

Plans in the drivable corridor, not painted lanes

02



Mixed Traffic

Cars, buses, trucks, 2W, autos, people, animals together

03



Irregular Road-User Behaviour

Reads intent from short-term motion history

04



Pedestrian & Cyclist Interaction

Acts on people entering the path, not the footpath

05



Wrong-Way / Informal Movement

Flags movement that enters the ego path

06



Animal & Temporary Obstacles

Plans around cattle, carts and work zones

07



Sudden Direction Changes

Re-evaluates every road user every frame

08



Dynamic Path Replanning

Keeps choosing a new safe path as the scene changes



DIFFERENTIATION & KEY VALUE PROPOSITION

Innovation and uniqueness of the solution

01

Ego-Path-Centric Risk Assessment

Judges the threat to the vehicle's own path — not mere proximity.

02

Indian Mixed-Traffic Awareness

Built for autos, two-wheelers, pedestrians and animals.

03

Short-Term Motion Understanding

Tracks each road user's motion to anticipate conflicts.

04

Adaptive Trajectory Generation

Samples kinematically feasible candidate paths every frame.

05

Lane-Independent Planning

Works where lane markings are faded, partial or absent.

06

Safety-Oriented Decision Making

GO → SLOW → STEER → BRAKE → NO SAFE PATH, each with a reason.

07

Dynamic Replanning

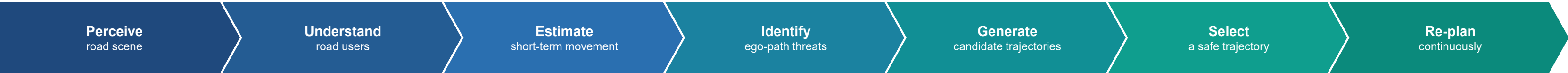
Re-plans continuously as traffic and obstacles change.

08

Scalable Multi-Sensor Roadmap

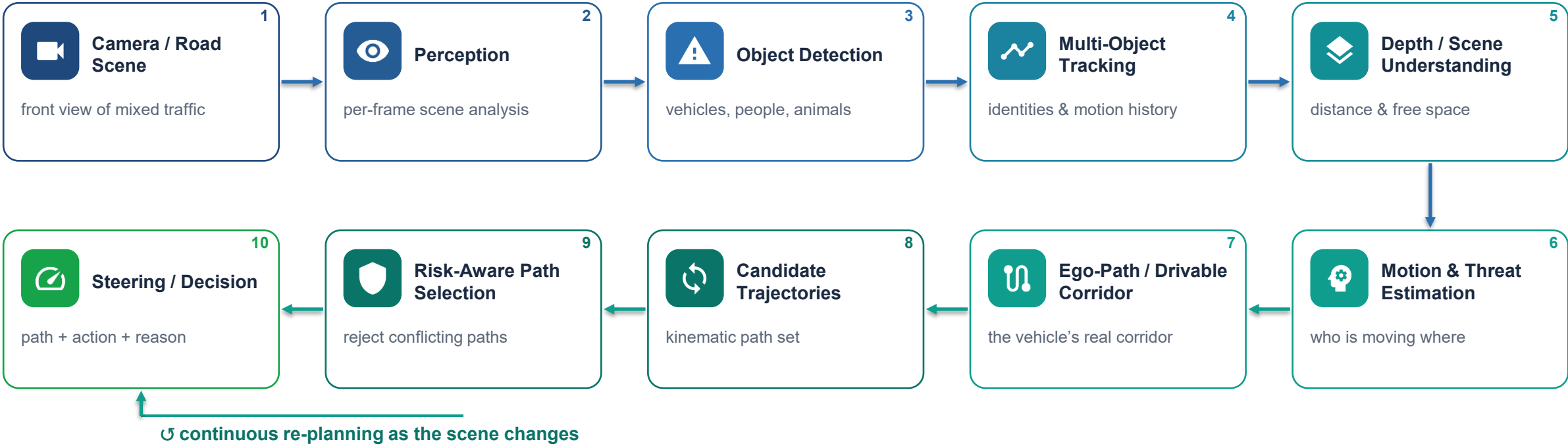
Camera-first today; radar, LiDAR and vehicle-state inputs later.

CORE
IDEA



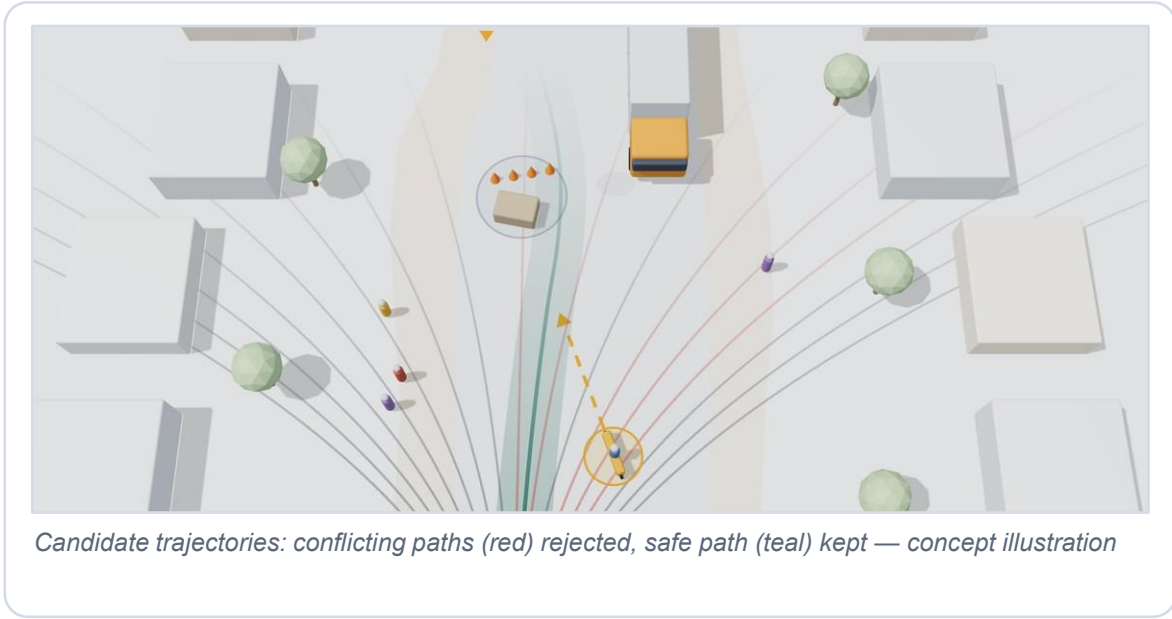
METHODOLOGY & ARCHITECTURE
Methodology and process for implementation (Flow Charts/Images/ working prototype)

SENSE & UNDERSTAND → row 1 **PLAN & DECIDE** → row 2 (right to left)



DECISION LAYER — escalates with the threat to the ego path

GO	path clear, or traffic stays on its own side
SLOW DOWN	a road user is approaching the path
STEER	a safer offset trajectory exists
BRAKE	a road user is entering / in the path
NO SAFE PATH	every candidate path is blocked → stop



TECH STACK
Technologies to be used (e.g. programming languages, frameworks, hardware)

 Python core language	 OpenCV video & vision
 YOLO object detection	 PyTorch deep-learning runtime
 ByteTrack multi-object tracking	 Depth Anything V2 monocular depth
 NumPy planning & geometry	 CUDA GPU acceleration
 Flask demo web server	 HTML / CSS / JS review interface



FROM CAMERA VIEW TO DECISION

Actual output frames from the processed demo video — New BEL Road, Bengaluru (consecutive moments)

1 GO STRAIGHT



Oncoming car stays on its own side — ego path clear, keep going.

2 SLOW DOWN



Oncoming vehicle starts entering the ego path — threat approaching.

3 BRAKE



Pedestrian in the ego path — collision risk, brake.



BEFORE vs AFTER

Same instant — raw camera frame vs. PathSense output (Nandidurga Road, Bengaluru)



BEFORE — raw camera view



AFTER — PathSense: road users tracked, path clear → GO

CURRENT DEVELOPMENT

Camera-based proof-of-concept demonstrating perception, threat assessment and adaptive path decisions on mixed Indian-road scenes.

- Detects and tracks cars, buses, two-wheelers, auto-rickshaws, pedestrians and animals
- Separates traffic on its own side from threats entering the ego path
- Plans around obstacles with candidate trajectories
- Explains every decision and replays key moments

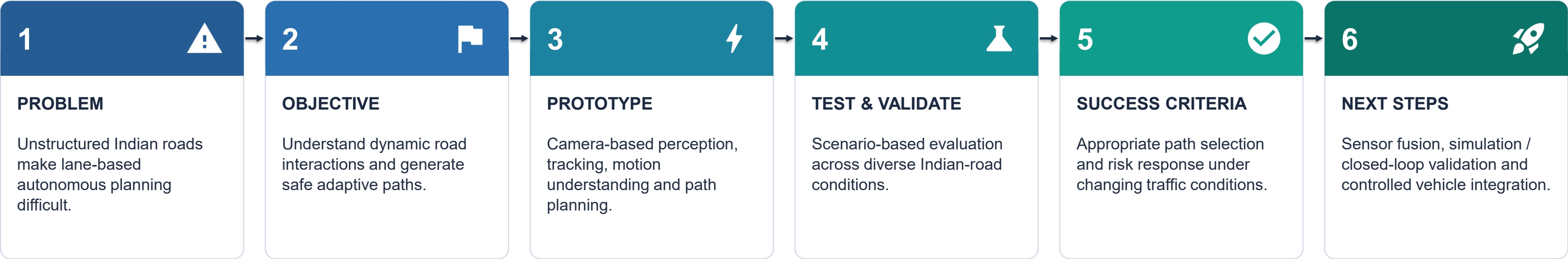


Source code

github.com/MalyalaKarthik66/pathsense-prototype

Frames are unedited pipeline output, cropped to the camera view. Source footage: L. Shyamal, Wikimedia Commons (CC0). Live demo shown separately.

PoC PROCESS — KEY STEPS
Define Problem & Objective · Set Success Criteria · Plan Scope & Resources · Build Prototype / Model · Test & Validate · Decide Next Steps



SCOPE & RESOURCES
Plan Scope & Resources

IN SCOPE — proof of concept

- Camera-based perception of mixed Indian traffic
- Multi-object tracking and short-term motion cues
- Ego-path threat assessment and candidate-path planning
- Explainable GO / SLOW / STEER / BRAKE decisions on recorded road video

LATER — pilot / production

- Vehicle actuation and closed-loop control
- Radar / LiDAR and vehicle-state fusion
- Functional-safety engineering (ISO 26262 / SOTIF)
- Controlled field trials at scale

RESOURCES

- Open pretrained models (YOLO, ByteTrack, Depth Anything V2)
- Openly licensed Indian road-scene videos
- Python / PyTorch stack on a laptop-class GPU
- Simulation tools for the pilot stage

FEASIBILITY
Analysis of the feasibility of the idea

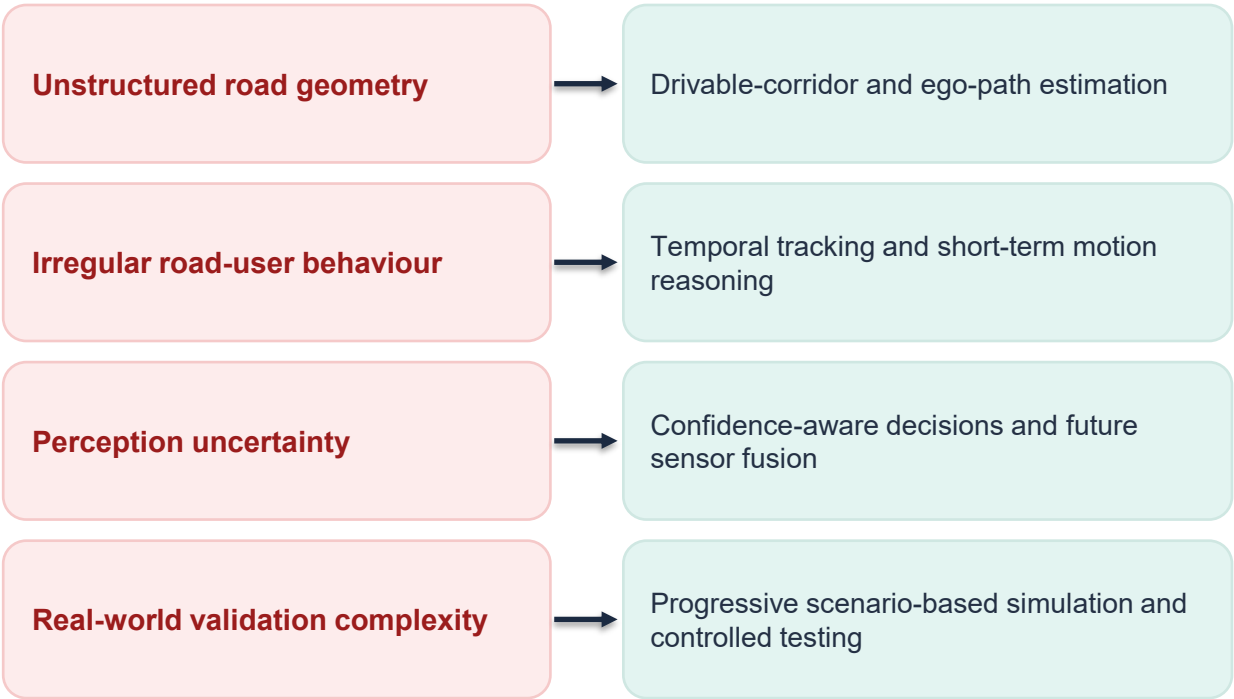
Technology: Established computer-vision, object-tracking, depth-estimation and trajectory-planning techniques provide a practical development base.

Implementation: Modular perception → prediction → planning architecture enables incremental development and validation.

Integration: Can progressively incorporate vehicle-state, radar and LiDAR inputs.

Validation: Scenario-based testing progresses from recorded road scenes to simulation and controlled field trials.

RISKS → MITIGATION
Potential challenges and risks · Strategies for overcoming these challenges



DATA-DRIVEN / OPERATIONAL VIABILITY

Data Foundation: road video, traffic observations, map / context information and future multi-sensor inputs.

Operational Value: adaptive navigation in mixed and unstructured road environments.

Scalability: prototype → pilot → production deployment pathway.

WHY IT MATTERS — India, 2022

4,61,312 road accidents **1,68,491** deaths

Source: MoRTH, Road Accidents in India 2022

PROTOTYPE → PILOT → PRODUCTION
Development roadmap

PROTOTYPE

Camera-based proof-of-concept for perception, tracking and adaptive path planning.

PILOT

Broader scenario validation, sensor expansion, simulation and controlled road trials.

PRODUCTION

Vehicle-grade multi-sensor perception, closed-loop validation and automotive safety engineering.

POTENTIAL COLLABORATIONS
Industry · academia · government


 Automotive OEMs & Tier-1 suppliers


 Research labs (IITs, IISc)


 Simulation-tool partners (e.g. MathWorks)


 Transport & road-safety departments


 Fleet & public-transport operators


 Mobility startups

 POTENTIAL IMPACTS — TARGET AUDIENCE
Potential impact on the target audience



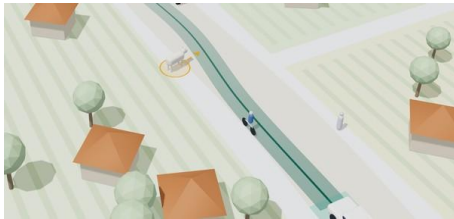
Passengers / Drivers
Safer interaction with unpredictable road users.



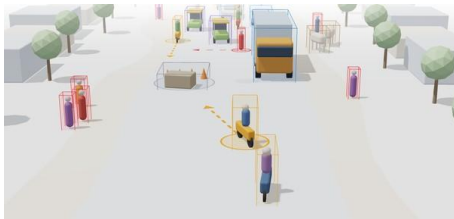
Urban Mobility
Adaptive navigation in dense mixed traffic.



Rural Mobility
Improved handling of roads without reliable lane markings.



Automotive / Mobility Industry
Foundation for Indian-road-aware autonomous systems.



Research & Academia
Platform for studying unstructured-road perception and planning.



concept illustrations

 BENEFITS
Benefits of the solution (social, economic, environmental, etc.)



SAFETY
Better recognition of potential ego-path conflicts.



ADAPTABILITY
Responds to changing traffic and road conditions.



SCALABILITY
Prototype → pilot → production pathway.



INDIAN-ROAD RELEVANCE
Designed around mixed and irregular traffic behaviour.

SOCIAL · ECONOMIC · ENVIRONMENTAL

- Social:** safer shared roads for pedestrians, cyclists and two-wheelers.
- Economic:** fewer conflicts for fleets and home-grown driver-assistance capability.
- Environmental:** smoother, anticipatory driving with fewer abrupt stops (expected).



Business Model Canvas



The Business Model Canvas — PathSense

Designed for:
PathSense

Designed by:
Ctrl Alt Elite

Date:
2026

Version:
Idea stage

Key Partners - Automotive OEMs - Mobility companies - Research institutions (IITs, IISc) - Sensor / perception providers - Simulation platforms (e.g. MathWorks) - Government / transport organisations	Key Activities - Perception development - Motion prediction - Path planning - Scenario validation - Sensor integration	Value Propositions - Adaptive planning for unstructured roads - Indian mixed-traffic awareness - Safety-oriented path decisions - Explainable: every decision has a reason - Scalable autonomous-navigation architecture	Customer Relationships - Co-development pilots - Integration & tuning support - Continuous model and scenario updates	Customer Segments - Automotive manufacturers - Autonomous mobility companies - Fleet operators - Research organisations - Smart mobility programmes
	Key Resources - AI / ML models - Indian road datasets - Simulation environments - Engineering team		Channels - Direct B2B partnerships - Pilot deployments with fleets - Research collaborations - Licensable SDK / API	
Cost Structure - Engineering R&D - Data collection & annotation - Compute & simulation infrastructure - Validation & safety engineering			Revenue Streams - Software licensing to OEMs - Pilot / integration contracts - Scenario-library & validation services - Support & update subscriptions	



GOVERNMENT / NATIONAL ALIGNMENT

Connect your solution with national missions



Digital India

How PathSense aligns

Software-first, camera-based mobility intelligence built on open digital tooling.



Smart Cities Mission

How PathSense aligns

Adaptive navigation for dense, mixed urban traffic and future smart-mobility services.



Atmanirbhar Bharat

How PathSense aligns

Indigenous autonomous-driving technology designed for Indian road conditions.



Swachh Bharat / Green India

How PathSense aligns

Smoother, anticipatory driving with fewer abrupt stops — an expected efficiency benefit.



AI for India 2.0

How PathSense aligns

Applied AI solving an India-specific mobility and road-safety problem.



National Road Safety (MoRTH)

How PathSense aligns

Earlier recognition of ego-path conflicts with pedestrians, cyclists and two-wheelers.



SUPPORTING RESEARCH & SOURCES

Details / Links of the reference and research work

AUTONOMOUS DRIVING

Paden et al., “A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles”, IEEE T-IV, 2016

arxiv.org/abs/1604.07446



OBJECT DETECTION

Redmon et al., “You Only Look Once: Unified, Real-Time Object Detection”, CVPR, 2016

arxiv.org/abs/1506.02640



MULTI-OBJECT TRACKING

Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box”, ECCV, 2022

arxiv.org/abs/2110.06864



MONOCULAR DEPTH

Yang et al., “Depth Anything V2”, NeurIPS, 2024

arxiv.org/abs/2406.09414



MOTION PREDICTION

Alahi et al., “Social LSTM: Human Trajectory Prediction in Crowded Spaces”, CVPR, 2016

openaccess.thecvf.com



INDIAN ROAD PERCEPTION

Varma et al., “IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments”, WACV, 2019

idd.insaan.iit.ac.in



OFFICIAL SOURCE

Smart India Hackathon 2026 — Problem Statement SIH26037: Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

sih.gov.in



ROAD-SAFETY DATA & FOOTAGE

MoRTH, “Road Accidents in India 2022”; demo road videos: L. Shyamal, Wikimedia Commons (CC0)

morth.nic.in · commons.wikimedia.org